

NSE Turnaround Stock Screener

Quantitative Momentum & Operational Recovery Framework

1 Project Overview

The **NSE Turnaround Stock Screener** is a high-performance quantitative tool designed to scan the National Stock Exchange (NSE) of India. Its primary objective is to identify high-quality companies that have experienced a transient financial setback but possess strong fundamentals indicating an immediate operational recovery.

By targeting the structural lag in market pricing efficiency, the screener highlights asymmetric risk-reward entry points before the broader market fully prices in the corporate turnaround.

2 The Core Strategy in Brief

The algorithm continuously monitors and filters companies matching a strict 4-quarter Net Income trajectory denoted as:

$$Q_{-3} \rightarrow Q_{-2} \rightarrow Q_{-1} \rightarrow Q_0$$

Quarter	Condition	Status	Strategic Significance
Q_{-3}	Net Income > 0	Profitable	Establishes historical baseline of healthy operations.
Q_{-2}	Net Income > 0	Profitable	Confirms consistency of the operational baseline.
Q_{-1}	Net Income ≤ 0	Loss-making	The "Setback" phase: transient deviation from trend.
Q_0	Net Income > 0	Profitable	The "Recovery" phase: verified return to baseline.

Core Thesis: This trajectory isolates fundamentally sound businesses that suffered an isolated, non-structural disruption (e.g., a one-time asset write-down, temporary supply shock, or localized R&D spike) but corrected course immediately.

3 Detailed Market Mechanics & Rationale

3.1 The Single-Quarter Deviation Anomaly

Standard value screeners search for long-term depressed assets, which often exposes investors to "value traps." Conversely, growth screeners favor persistent quarter-over-quarter expansion, which commands premium valuations.

This strategy occupies a unique mathematical middle ground: identifying highly efficient, cash-generating businesses experiencing a short-term volatility window. When a consistently profitable company prints a single-quarter net loss, institutional algorithms and panicked retail investors frequently trigger aggressive sell-offs, resulting in an overextended price correction.

3.2 Strategy Flowchart

Below is the full operational progression map identifying the exact transition from baseline metrics to the post-recovery pricing window:



3.3 Post-Earnings Announcement Drift (PEAD)

A core catalyst for this strategy is **Post-Earnings Announcement Drift (PEAD)**. Financial markets do not price new information instantaneously. Following the positive earnings surprise in Q_0 :

1. Institutional pools require days or weeks to accumulate or rebuild significant positions to minimize market impact.
2. Analysts slowly revise their forward guidance models.

This structural delay creates a repeatable window of opportunity where the stock trades at a value discount despite verified fundamental recovery.

4 Quantitative Risk & Valuation Filtering

To prevent selecting false positives or structurally illiquid businesses, the screener applies a secondary layer of risk filters:

4.1 Solvency Check (Quick Ratio)

Ensures the target entity maintains sufficient short-term liquidity to manage operational shocks without encountering insolvency risk.

$$\text{Quick Ratio} = \frac{\text{Current Assets} - \text{Inventory}}{\text{Current Liabilities}} \quad (1)$$

A **Quick Ratio** > 1.0 is strictly required to verify short-term financial resilience.

4.2 Valuation & Growth Metrics

The screener extracts the trailing **Price-to-Earnings (P/E)** and **Price/Earnings-to-Growth (PEG)** ratios to quantify the mispricing relative to forward earnings velocity:

$$\text{PEG Ratio} = \frac{\text{P/E Ratio}}{\text{Annual EPS Growth Rate}} \quad (2)$$

Lower PEG ratios confirm that despite the recovery bounce, the stock remains fundamentally undervalued relative to its historical and projected growth trajectory.